

CA/Browser Forum Tracker

Part 5 - Revocation & Transparency

Compiled by the Spaced Research Ledger

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Revocation and transparency are the two mechanisms that let the web recover from a certificate that should not exist. Revocation withdraws trust from a certificate already issued. Transparency makes every issued certificate publicly visible, so that a bad one can be found at all. Section 1 covers revocation, where the industry's answer has shifted from fixing the mechanism to routing around it. Section 2 covers Certificate Transparency, which is undergoing its first architectural change since it was standardized.

1. Revocation Mechanisms and Timelines

Revocation is the part of the web's certificate system that never worked well. The clearest evidence of that judgment is that the largest certificate authority in the world switched off its status responder and replaced it with certificates too short-lived to need one.

Recent Developments

Let's Encrypt ended its Online Certificate Status Protocol service in August 2025 [1]. It had stopped including responder URLs in certificates more than ninety days earlier, so every certificate carrying one had expired before the responder went dark [1]. The stated reason was privacy, because the protocol tells the responder which site a user is visiting [2].

The scale of what was removed is worth stating plainly. At its peak the service answered roughly 340 billion requests per month, with more than 140,000 requests per second handled by the content delivery network [3]. The intent to replace the protocol with revocation lists was announced in July 2024, more than a year before the shutdown [4].

The replacement is not a better status protocol. It is a shorter certificate. Let's Encrypt issued its first six-day certificate on 20 February 2025 [5]. Those certificates carry neither responder nor revocation list URLs, because the window in which a compromised key is useful is short enough that revocation is treated as unnecessary [6].

Current State

Browser-side revocation summaries diverge sharply in ambition. Firefox began production use of CRLite in August 2025, covering 99.84% of certificates visible in transparency logs [7]. It arrives as roughly 300 kB of daily updates and is queried locally with no network lookup [7]. Chrome's CRLSets carry about 1% of all revocations, roughly 35,000 out of four million [8].

The efficiency gap is structural rather than a matter of effort. CRLite encodes its set in about 300 kB per day against roughly 600 kB for a recent CRLSet, so it delivers complete coverage in half the bandwidth [9]. It uses

about 6.6 bits per revocation against roughly 110 bits for a CRLSet, because a filter cascade encodes membership rather than listing entries [10].

The two designs answer different questions. Mozilla pairs OneCRL for intermediate certificate revocations with CRLite for end-entity revocations. Chrome's CRLSets are a selective list aimed at high-impact and emergency revocations rather than comprehensive coverage [11].

Policy is tightening around the timelines themselves. Mozilla requires certificate authorities to develop, maintain and annually test comprehensive mass revocation plans, with third-party assessment from June 2025 audits [12]. It states that there are no exceptions to the Forum's revocation timelines [12]. That policy followed multiple delayed-revocation incidents, and version 3.0 added a requirement for proactive subscriber warnings about revocation timelines and contractual language mandating timely cooperation [13].

Unresolved Conflicts

The Forum's own revocation deadlines are debated but unsettled. Working group discussions in 2025 proposed splitting one requirement into separate triggers. The split would distinguish non-compliance with the Baseline Requirements from deviations against an authority's own policy statement, with a 90-day timeline for the latter [14]. The related ballots remained in draft as of December 2025 [14]. Until one is balloted, the existing deadlines stand unchanged, and the argument for a slower track for less severe problems remains an argument rather than a rule.

2. Certificate Transparency

Certificate Transparency is moving from its original append-only log design to a static, tiled architecture that operators can serve from ordinary object storage. Chrome is driving that transition and has staged it as an admission process rather than a cutover.

Recent Developments

Chrome's log policy now accepts logs conforming either to the original RFC 6962 design or to the static specification, and neither has been retired [15]. The change took effect in Chrome 134, released 4 March, with operators able to submit static logs for inclusion from 1 April [16].

Chrome then went further than admitting the new design. It removed the requirement to include at least one log of the original type. From 15 April 2026 a certificate may carry signed timestamps drawn exclusively from static logs [17]. Chrome will also remove support for timestamps delivered by status responder in Chrome 148 [18].

The bar for the new logs is high. Chrome will only consider static logs with a maximum merge delay of ten seconds or less, far tighter than the day-long norm of the original design. Applications are accepted only from operators who have already applied with a log of the original type accepting the same certificates.

Current State

Logging obligations are now near universal. Chrome Root Program Policy version 1.8 requires certificate authorities to log all TLS server authentication precertificates to at least one qualified or usable log, effective 15 June 2026 [19].

The implementation ecosystem is active rather than settled. The static specification and several tiled log implementations continue development, including support for temporal sharding by certificate expiry window [20]. Chrome's policy defines a lifecycle for logs running through pending, qualified, usable, read-only, retired and rejected states [21]. The monitoring side is also maintained, with the crt.sh project carrying repository updates as recent as September 2026 [22].

One operational incident is on record. The Tuscolo2026h2 static log suffered a thirty-minute degradation of its write path on 9 August 2026, caused by a traffic spike and a lock starvation bug [23]. All entries still met the sixty-second merge delay [23].

Beyond the current architecture, work continues on a design that merges logging into issuance itself. The IETF draft for Merkle Tree Certificates reached version 05 in July 2026 [24].

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to four systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

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